

Deepak Mallampati

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PROFESSIONAL SUMMARY

LLM platform engineer with 2.5+ years building production RAG systems, agentic reasoning workflows, and LLM-powered security-adjacent applications. Proven track record delivering measurable AI outcomes: ~35% retrieval precision improvement, >30% hallucination reduction via structured agent reasoning, and 15-20% recall gains on high-risk prediction tasks. Deep hands-on experience with LangChain, LlamaIndex, PyTorch, Hugging Face Transformers, and evaluation pipelines. Comfortable owning end-to-end delivery from prompts and models to APIs, Docker deployment, and production observability - with a first-principles mindset for breaking down complex AI and data challenges.

CORE COMPETENCIES

LLM Application Development • RAG Pipelines • Agentic AI Workflows • LangChain • LlamaIndex • PyTorch • Hugging Face Transformers • Prompt Engineering • LLM Evaluation Pipelines • Vector Search & Embeddings • ReAct / Multi-Agent Orchestration • FastAPI • Docker • Python • TypeScript • CI/CD • Responsible AI & Guardrails

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer - Healthcare Technology & AI Consulting

- Built production **RAG systems** for clinical knowledge retrieval and documentation search; implemented recursive semantic chunking and transformer-based embeddings; improved contextual retrieval precision by ~35%; retrieval-grounded response alignment exceeded 90% in evaluation.
- Developed **agentic LLM workflows** for multi-step healthcare queries with structured reasoning, tool discipline, and grounding rules - reducing agent response instability by ~25% and hallucinations by >30%; directly applicable to reasoning-driven security automation.
- Built **LLM evaluation pipelines** with structured logging, audit trails, and system-limitation documentation to track model performance, surface regressions, and maintain responsible AI standards.
- Shipped **FastAPI/Flask** RESTful inference services containerized with **Docker**; load simulation and observability tooling reduced post-deployment defects by ~30%.
- Built preprocessing pipelines (Pandas, NumPy) for EHR data; raised dataset reliability above 98% and cut downstream model instability by ~40%.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Oil & Gas ERP

- Optimized SQL and **T-SQL stored procedures**; reduced query execution time ~20% and data retrieval latency ~25%.
- Built **CI/CD pipelines with Jenkins** for schema updates and stored procedure deployments; reduced deployment errors >30%.
- Designed performance dashboards using SQL Server DMVs and **Grafana**; cut incident recurrence ~25%.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Nonprofit Video Analytics & NLP

- Built **transformer-based video summarization** system across 5,000+ sessions; reduced manual review time by 60-70%.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs ~30% and raised contextual accuracy >85%.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline (Intake → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation; ANN similarity search for duplicate detection; explainable risk scoring with audit-ready reasoning traces - directly analogous to AI security agent reasoning workflows.

Python, LangChain, LlamaIndex, RAG, multi-agent orchestration, ReAct patterns, vector search

Manga Lens - Chrome Extension for Real-Time AI Translation shipped

Full-stack browser extension (Manifest V3, TypeScript, service workers) integrating **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama local). Multi-section capture pipeline, 7-day translation cache, per-domain configs for 29 sites - demonstrates production TypeScript API integration and multi-provider LLM orchestration.

TypeScript, Manifest V3, REST APIs, LLM orchestration, service workers

Agentic Pixel Character Synthesis & Animation Engine

LLM-based agent orchestrator that decomposes high-level prompts into generation tasks; Stable Diffusion fine-tuning + LoRA for identity consistency; ControlNet + OpenPose for pose conditioning.

PyTorch, Hugging Face Diffusers, FastAPI, Docker, LoRA fine-tuning, LLM orchestration

Dream Decoder - AI-Powered Multimodal Platform

FastAPI + React/TypeScript/Vite app with **intermediate structured prompt transformation layers** improving contextual alignment ~**30%** and first-pass image success ~**25-30%**.

FastAPI, React, TypeScript, LLM APIs, prompt engineering, multimodal orchestration

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

LLM & GenAI: LLM application development, RAG, agentic workflows, ReAct patterns, prompt engineering, LLM evaluation, guardrails, grounding, structured outputs, vector search, embeddings, responsible AI

ML Frameworks: PyTorch, Hugging Face Transformers, scikit-learn, XGBoost, LangChain, LlamaIndex

Languages: Python, TypeScript, SQL (T-SQL, PostgreSQL), FastAPI, Flask, React

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, RESTful APIs, cloud-ready deployment, observability/logging

Work Authorization: F-1 OPT - authorized to work in the United States now